Response to Reviewer 1

"The Seifert–van Kampen Theorem via Computational Paths"

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We thank the reviewer for the careful and constructive report. We retained the paper's structure and addressed each identified location directly. The Lean development now includes proofs of both glue-naturality forms, the obstruction to an amalgamation-only SVK target, the full-target theorem for the global rewrite quotient, the low-degree π_0 tail of the fibration sequence, and a general sphere loop-quotient theorem. It also includes the unconditional space-level circle realization theorem described below.

As a final claim-level audit, we corrected two possible over-readings in the extended-feature sections. The revised manuscript restricts the Eckmann–Hilton theorem to the explicitly defined π_2, labels the $n \geq 3$ branches of `HigherHomotopy.PiN` as current unit placeholders, and presents the fibration construction as a low-degree sequence with pointed adjacent-composition laws rather than a full image-kernel exactness theorem. We also moved the HoTT comparison table inside Section 10.1 and made the formalization architecture explicitly Table 2. For navigation, the principal claim repairs are in Sections 2.5, 7.1, 7.5, 9.1, and 10.1, with the declaration-level mapping in Appendix A.

The revision also makes the quotient being classified explicit in every nontrivial application. The relation generated by all constructors of the global `Path.Step` system makes each `PathRwQuot` loop fiber contractible.

To obtain a non-collapsing fundamental-group theorem without relabeling a normal-form quotient, we added `PresentedFundamentalGroup.lean`. A presented path space has proof-relevant generating edges; raw paths are freely generated by reflexivity, reversal, and composition; and homotopy is generated only afterward from named relators and the groupoid laws. Its fundamental group is the automorphism group of the basepoint in the resulting strict groupoid. The new theorem `presentedSeifertVanKampenGroupEquiv` constructs encode and decode, proves both round trips, and records preservation of multiplication, identity, and inversion:

$$\pi_1^{\mathrm{pres}}(\mathcal{P}(G_1, G_2; H), *) \cong G_1 *_H G_2.$$

We instantiate it for the figure-eight and separately prove a nontrivial generator. The circle presentation is likewise proved equivalent to $\mathbb{Z}$. At the fundamental-group level, we additionally prove that $\mathbb{R} \to \mathsf{AddCircle}\,1$ is a covering map, classify Mathlib's topological fundamental group of the circle by winding, and prove the group equivalence `presentedCircleTopologicalPiOneGroupEquiv`. A general geometric realization is also constructed from the nerve, and `Presented.Realization.hoNerveIso` recovers the original presented groupoid as its simplicial homotopy category. The final comparison with the topological fundamental groupoid is now represented by an explicit canonical functor, whose identity and composition laws, essential surjectivity, fullness, and faithfulness are proved. We descend degeneracy-aware open core-face stars through the realization quotient, prove that they cover, partition each preimage into disjoint lifted sheets, construct the sheet homeomorphisms, and prove that the under-category projection is a Mathlib covering map. Path and homotopy lifting over its contractible total space then provide the explicit hom-set inverse and the public equivalence `topologicalFundamentalGroupoidEquivalence`. This closes the general nerve-realization/fundamental-groupoid gap that the previous draft listed under both Limitations and Future Work. Separately, `CircleNerveAmbient.lean` now proves the concrete ambient statement

$$\mathsf{CircleNerveRealization} \simeq_h \mathsf{TopologicalCircle}.$$

The maps `circleNerveToAmbient` and `circleAmbientToNerve` have an exact right inverse, while a face- and degeneracy-compatible simplexwise join homotopy descends through the realization colimit to prove the other inverse up to homotopy. Thus the circle case is no longer an ambient-space limitation; the analogous comparison for arbitrary concrete CW presentations remains Future Work.

We also completed the topological structure carried by the computational paths themselves. The new declaration `unconditionalTopologicalGroupoidCertificate` packages continuous source, target, identity, inverse, and composition together with the endpoint, unit, inverse, and associativity laws on quotient classes. It is specialized to all presented realizations by `continuousPresentedTopologicalGroupoidCertificate`, and transported along any supplied ambient homotopy equivalence by `continuousAmbientTopologicalGroupoidCertificate`. The unconditional composition domain has the final topology induced from composable representatives. We state separately that continuity for the ordinary product/subspace topology retains the explicit `ProductQuotientCompatibility` hypothesis, since products of quotient maps are not quotient maps in arbitrary topological spaces.

Because these additions change where the paper places its weight, we also added two short remarks that state their status plainly. Remark 2.10 records that the presented path space is the standard combinatorial presentation of a groupoid, citing Higgins and Brown, so that the reader can see it is not an object introduced to make a theorem succeed; Brown states Seifert–van Kampen for fundamental groupoids in exactly this form. Remark 2.7 gives the precise mechanism of the global-rule collapse and reads it as a constraint on the design space rather than as a defect of one implementation.

Overall remarks

Reviewer comment. *The assumption bookkeeping is not applied consistently, and the reader must consult several separated tables and appendices to determine what is proved.*

Response. We consolidated the bookkeeping in four coordinated places: the main presented theorem prints its algebraic data directly, the separate global-rule schema prints all encode typeclasses in its statement, Section 8 contains one manifest for those conditional parameters, and Appendix A gives, for every headline result, the exact Lean name, module, object classified, and status ("presented fundamental group", "global-rule quotient", "conditional", "completed presentation", or "open"). The text explains that the amalgamation-only relation preserves word length and cites the formal impossibility theorems for that target.

Reviewer comment. *Claims about decidability and the "36 axioms" count are weaker than the headline presentation suggests.*

Response. Both claims were corrected. Arbitrary RwEq is no longer described as decidable without qualification: the paper now states that expression-level termination and confluence are proved, while path-level decidability requires an explicitly supplied `CompleteStepSystem`. The "36 axioms" headline and optional axiom-import inventory were removed. The source contains zero custom `axiom` declarations, as checked by `scripts/check_invariants.py` and `paper/svk/check_stats.py`.

Reviewer comment. *A key naturality lemma is asserted without proof.*

Response. We added a complete derivation in the paper, together with the two Lean theorems `Pushout.glue_natural_square_rweq` and `Pushout.glue_natural_rearranged_rweq`. Both are proved in the module `Path/CompPath/PushoutCompPath.lean`.

Detailed comments

Reviewer comment. *p. 2, Section 1, item 2: "$\sim$ is decidable via normalization" conflicts with the metatheory table.*

Response. Item 2 states that the normalization function is a sound invariant, not an unconditional complete characterization. It identifies the proved abstract-expression fragment and points to `CompleteStepSystem.decideRwEq` for conditional decidability.

Reviewer comment. *p. 10, Section 2.4: confluence is marked "typeclass", but it is not stated who discharges it.*

Response. The table now distinguishes two levels. The library discharges expression termination and confluence through `instHasTerminationProp` and `instHasConfluencePropExpr`. A caller must supply `HasJoinOfRw`/`HasConfluenceProp` at the trace-carrying `Path` level; no global instance is claimed.

Reviewer comment. *p. 16, Lemma 3.4: no proof is given; add a proof or Lean pointer.*

Response. The added proof sets $L = \mathsf{inl}_*(f_*(p))$ and $R = \mathsf{glue}(c_1) \cdot \mathsf{inr}_*(g_*(p))$, uses the stored witness $L^{-1} \cdot R \sim \mathsf{glue}(c_2)$, and then applies left congruence, associativity, inverse cancellation, and the unit law. The paper cites `glue_natural_square_rweq`.

Reviewer comment. *p. 16, Corollary 3.5: state explicitly that it follows by right-multiplying by* $\mathsf{glue}(c_2)^{-1}$.

Response. The proof now says exactly this, including reassociation and inverse cancellation. The corresponding Lean theorem is `glue_natural_rearranged_rweq`.

Reviewer comment. *p. 21, Section 5.1: Theorem 5.1 gives no indication that it is conditional.*

Response. The revised Section 5 now begins with an unconditional theorem for presented computational path spaces. Its raw paths and generated homotopy are defined before encode, and Lean theorem `presentedSeifertVanKampenGroupEquiv` constructs both maps, proves both round trips, and packages the group homomorphism laws. Its hypotheses `GroupLaws` and `AmalgamationLaws` are explicit algebraic structures; the latter includes the source-group laws and two `PreservesGroupOps` certificates. None of these are encode assumptions. For the circle, the separate theorem `presentedCircleTopologicalPiOneGroupEquiv` identifies the presented fundamental group with Mathlib's topological `FundamentalGroup (AddCircle 1) 0`.

The separate theorem for the global-rule `PathRwQuot` is titled "Conditional global-rule SVK schema", and all four required typeclasses are part of its statement. We also distinguish the full target, which includes free-group reduction, from the amalgamation-only target. Lean proves that round-trip and bijectivity interfaces for the latter are impossible: `hasPushoutSVKEncodeDecode_impossible` and `hasPushoutSVKDecodeAmalgBijective_` `impossible`. Separately, `seifertVanKampenFullEquiv_collapsed` proves the full-target equivalence for the global rewrite quotient and is explicitly labeled degenerate. The displayed full-target definition now reproduces the six actual `FreeGroupStep` constructors—adjacent combination on both sides, zero removal on both sides, and left/right contextual congruence—and the explicit reflexive, amalgamation, free-group, symmetric, and transitive constructors of `FullAmalgEquiv`. As a final transcription cleanup, the displayed `Step` codomain is now `Type(u+1)`, and the displayed word-operation function is now the noncomputable `inverse` used by `PushoutPaths.lean`. The completed-expression theorem `scopedSeifertVanKampenEquiv` remains as a separate normal-form result rather than serving as the fundamental-group theorem.

Reviewer comment. *p. 27, Table 1: the sphere row says "SVK: Yes", while Appendix A lists no dependency for S^2.*

Response. We added the general Lean theorem `sphereN_piOne_equiv_unit`, specialized from `pathRwQuotLoopEquivUnit`. Table 1 and Appendix A agree: this global-rule loop-quotient result has no SVK encode/decode dependency and follows from the universal quotient collapse.

Reviewer comment. *p. 36, Definition 7.6: the long exact sequence is truncated at $\to 1$; should it continue through π_0?*

Response. Yes. The display continues

$$\pi_1(B) \to \pi_0(F) \to \pi_0(E) \to \pi_0(B).$$

The revised subsection is now titled "Low-degree fibration sequence". The Lean development defines `boundaryPi0`, `fiberPi0ToTotalPi0`, and `totalPi0ToBasePi0`, and packages them with the pointed adjacent-composition laws in `LowDegreeFibrationSequence`; the constructor is `lowDegreeFibrationSequence`. It does not claim full image-kernel exactness for the displayed sequence; that strengthening remains future work. The paper explains that a terminal 1 is shorthand only under path-connectedness assumptions (revised Section 7.5).

Reviewer comment. *p. 5, related work: add Agda/HoTT–Agda SVK developments and Lean Mathlib's fundamental groupoid material.*

Response. Section 1.2 discusses the machine-checked `VanKampen` development in HoTT–Agda and Mathlib's classical fundamental groupoid/fundamental group modules, and explains how their objects differ from explicit equality-trace quotients.

Reviewer comment. *No repository link is provided.*

Response. The abstract and a dedicated Code Availability subsection now cite the exact Zenodo version 0.4.2 matching this revision: `https://doi.org/10.5281/zenodo.21779801`. It contains the 31 paper entry modules and their 61 transitive local dependencies (92 Lean files total), including the topological computational path groupoid package, `CircleNerveAmbient.lean`, the pinned toolchain and Lake metadata, theorem manifest, license, invariant checker, and the corrected manuscript and response sources. Its archive SHA-256 is recorded in the Zenodo metadata. The concept DOI for the archival series is `https://doi.org/10.5281/zenodo.21696874`; the corresponding source release is `https://github.com/Arthur742Ramos/ComputationalPathsLean/releases/tag/v0.4.2`, and the repository is `https://github.com/Arthur742Ramos/ComputationalPathsLean`.

Reviewer comment. *p. 44, Section 8.6 Part C: optional axioms are excluded from the headline count and should be flagged next to each theorem.*

Response. The source contains no such optional axiom files, so Part C and the "36 axioms" count were removed rather than merely relocated. Each application is labeled locally as presented fundamental group, global-rule/proved, conditional, completed presentation, or open, and Appendix A repeats the same status.

Reviewer comment. *p. 36, path tactics: state what fraction of theorems rely on them.*

Response. We added a reproducible lexical count. None of 20,389 `theorem/lemma` declarations directly invokes a `path\_*` tactic. Across all 54,249 declarations counted, including definitions, instances, and examples, 11 do so (approximately 0.020%); in the thirty-one paper modules, one of 1,465 declarations does so (0.068%). The paper therefore characterizes these tactics as convenience wrappers rather than a mature dominant automation layer.

Reviewer comment. *The seventh line of the table on p. 51 does not express Theorem 6.8.*

Response. The Klein-bottle row and the corresponding theorem now name the declaration `kleinBottleScopedPiOneEquivIntProd`, and identify its source as the scoped expression quotient in `ClassicalPresentationsScoped.lean`. Both places state that the quotient is classified by the $\mathbb{Z} \rtimes \mathbb{Z}$ normal form and do not identify it with the global-rule `PathRwQuot`.

Minor corrections

- The structure paragraph now includes Section 9 before Section 10.
- The decode subsection now begins "For pushouts, the decode map converts a word to a loop", correcting the punctuation.
- The bullets in Example 6.12 begin on lines separate from the heading.
- "Part C" was removed with the optional-axiom inventory, eliminating the punctuation issue.
- Future Work now spells out the octonionic sequence $S^7 \to S^{15} \to S^8$.